Approach to patients with coma

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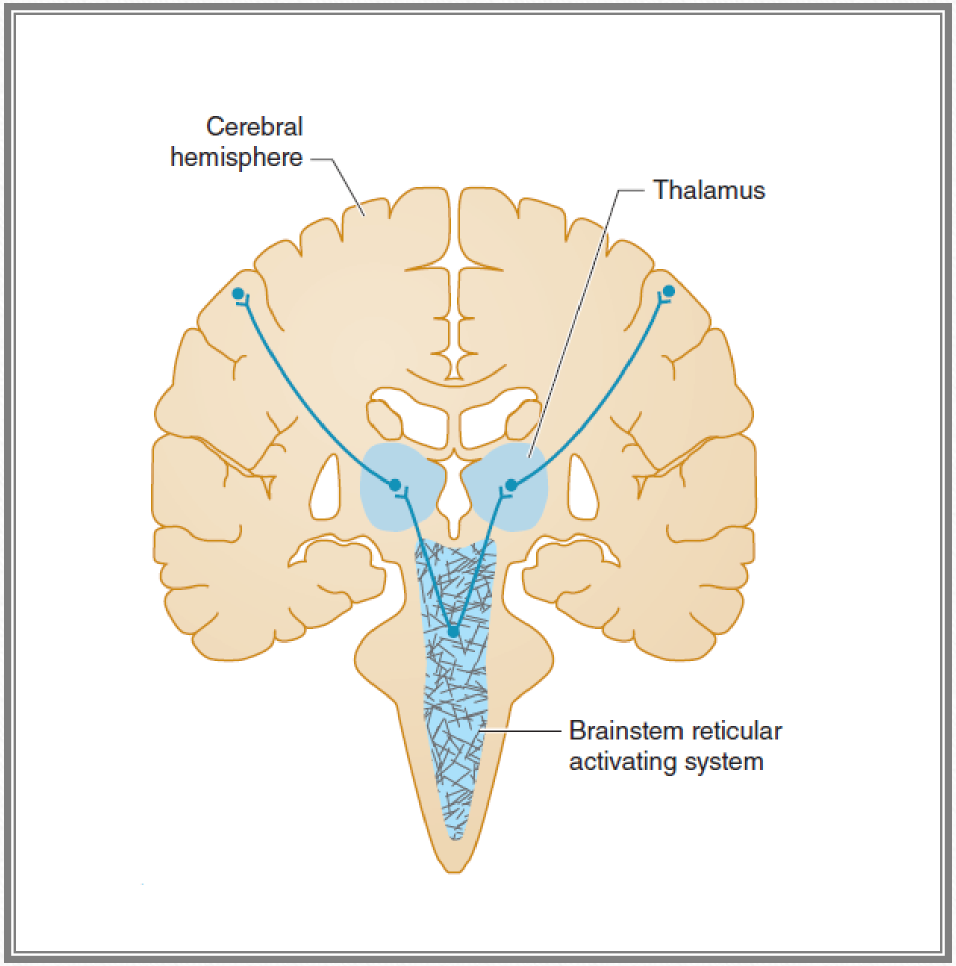
Definition

- Coma is a sleep-like state with no purposeful response to the environment and from which the patient cannot be aroused.
- The eyes are closed and do not open spontaneously.
- The patient does not speak, and there is no purposeful movement of the face or limbs. Verbal stimulation produces no response.
- Painful stimulation may produce no response or nonpurposeful reflex movements mediated through spinal cord or brainstem pathways.
- It results from a disturbance in the function of either the **brainstem reticular activating system(RAS)** above the mid pons or of **both cerebral hemispheres.**

RAS



Cerebral hemisphere
Thalamus
Brainstem reticular activating system

Approach to Diagnosis

- The approach to diagnosis of the comatose patient consists first of emergency measures to stabilize the patient and treat presumptively certain life-threatening disorders, followed by efforts to establish an etiologic diagnosis.

Emergency Management

1. ABC:

a) Airway
b) Breathing: Check for cyanosis, respiratory rate, presence of breath sound on auscultation, pulse oximetry, and arterial blood gas study.
c) Circulation: Pulse rate and blood pressure.

2. Insert an intravenous catheter and withdraw blood for determination of

a) Serum glucose
b) Electrolytes
c) Hepatic and renal function tests
d) Prothrombin time, partial thromboplastin time
e) Complete blood count
f) Drug screen.

3. Begin the routine combination of **dextrose**, **thiamine**, and **naloxone**.

- Every comatose patient should be given 25 g of **dextrose** intravenously, typically as 50 mL of a 50% dextrose solution, to treat possible hypoglycemic coma.

- Because administration of dextrose alone may precipitate or worsen Wernicke encephalopathy in thiamine-deficient patients, all comatose patients should also receive 100 mg of **thiamine** by the intravenous route.

- To treat possible opiate overdose, the opiate antagonist **naloxone**, 0.4 to 1.2 mg intravenously, should also be administered routinely to comatose patients.
- The benzodiazepine antagonist **flumazenil**, 1 to 10 mg intravenously, may be useful when benzodiazepine overdose contributes to coma.
- Flumazenil should not be used in patients with a history of seizures, chronic benzodiazepine abuse, or suspected co-ingestion of tri- or tetracyclic antidepressants (suspected if the electrocardiogram (ECG) shows sinus tachycardia at a rate of 130/min or more, QTc interval greater than 0.5 seconds, and QRS duration greater than 0.1 seconds).

4. Withdraw arterial blood for blood gas and pH determinations.

- These studies can provide clues to metabolic causes of coma.

5. Institute treatment for seizures, if present.

- Persistent or recurrent seizures in a comatose patient should be considered to represent status epilepticus and treated accordingly.

Emergency Management of the Comatose Patient

Immediately	Next	Later
Ensure adequacy of airway, ventilation, and circulation	If signs of meningeal irritation are present (see Figure 1-5), perform LP to rule out meningitis. Obtain a history if possible	ECG
Draw blood for serum glucose, electrolytes, liver and renal function tests, PT, PTT, and CBC	Perform detailed general physical and neurologic examination	Correct hyper- or hypothermia
Start IV and administer 25 g of dextrose, 100 mg of thiamine, and 0.4-1.2 mg of naloxone IV	Order CT scan of head if history or findings suggest structural lesion or subarachnoid hemorrhage	Correct severe acid–base and electrolyte abnormalities
Draw blood for arterial blood gas determinations		Chest x-ray
Treat seizures (see Chapter 12, Seizures & Syncope)		Blood and urine toxicology studies; EEG

Abbreviations: CBC, complete blood count; IV, intravenous; LP, lumbar puncture; PT, prothrombin time; PTT, partial thromboplastin time.

History and Examination

1. A **sudden onset** of coma suggests a vascular origin, especially a brainstem stroke or subarachnoid hemorrhage.
2. **Rapid progression** from hemispheric signs, such as hemiparesis, hemisensory deficit, or aphasia, to coma within minutes to hours is characteristic of intracerebral hemorrhage.
3. A **more protracted course** leading to coma (days to a week or more) is seen with tumor, abscess, or chronic subdural hematoma.
4. Coma preceded by a **confusional state or agitated delirium**, without lateralizing signs or symptoms, is probably due to a **metabolic derangement** or infection (meningitis or encephalitis).

GENERAL PHYSI CAL EXAMINATION

- **Signs of Trauma**
- Inspection of the head may reveal signs of **basilar skull fracture:**

a) **Raccoon eyes**

b) **Battle sign**

c) **Hemotympanum**

d) **Cerebrospinal fluid (CSF) rhinorrhea or otorrhea**

- Palpation of the head may demonstrate a **depressed skull fracture** or **swelling of soft tissues** at the site of trauma.

Raccoon eyes and Battle sign

- **Blood Pressure**

- **High BP** may reflect a long-standing hypertension, which predisposes to intracerebral hemorrhage or stroke.
- It could be due to hypertensive encephalopathy (BP exceeds 250/150 mm Hg in chronically hypertensive patients; it may be lower in children or after acute elevation of blood pressure in previously normotensive patients (e.g. in acute renal failure).
- Elevated blood pressure may also be a consequence of the process causing the coma, as in intracerebral or subarachnoid hemorrhage or, rarely, brainstem stroke.
- If **BP is low** or undetected causes of shock should be considered.

- **Temperature**

1. Hypothermia:

a) Ethanol
b) Sedative drugs (High dose benzodiazepine)
c) Hypoglycemia
d) Wernicke encephalopathy
e) Hepatic encephalopathy
f) Myxedema

1. Hyperthermia:

a) Status epilepticus
b) Meningitis or encephalitis
c) Anticholinergic drug intoxication (e.g. TCA)
d) Neuroleptic malignant syndrome
e) Thyrotoxic crisis.
f) Heat stroke
g) Malignant hyperthermia related to inhalational anesthetics
h) Pontine hemorrhage
i) Certain hypothalamic lesions.

- Patterns of Breathing

A. **Cheyne- Stokes respiration**—diffuse forebrain damage.
B. **Central neurogenic hyperventilation**—lesions of low midbrain ventral to aqueduct of Sylvius and of upper pons ventral to the fourth ventricle.
C. **Apneusis**—dorsolateral tegmental lesion of middle and caudal pons.
D. **Cluster breathing**—lower pontine tegmental lesion.
E. **Ataxic breathing**—lesion of the reticular formation of the dorsomedial part of the medulla.
F. **Gasping**.

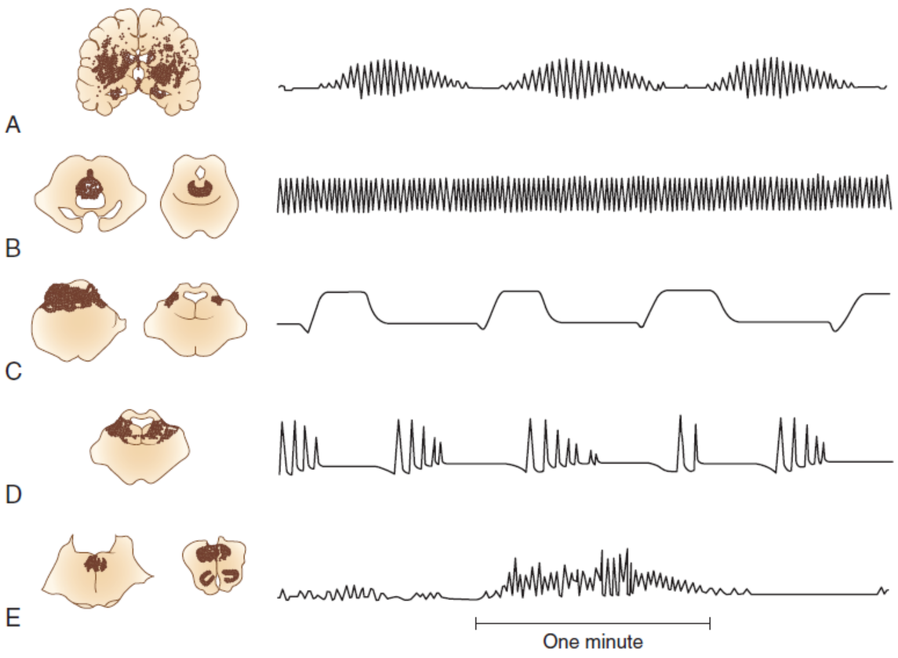
A. Cheyne-Stokes respiration

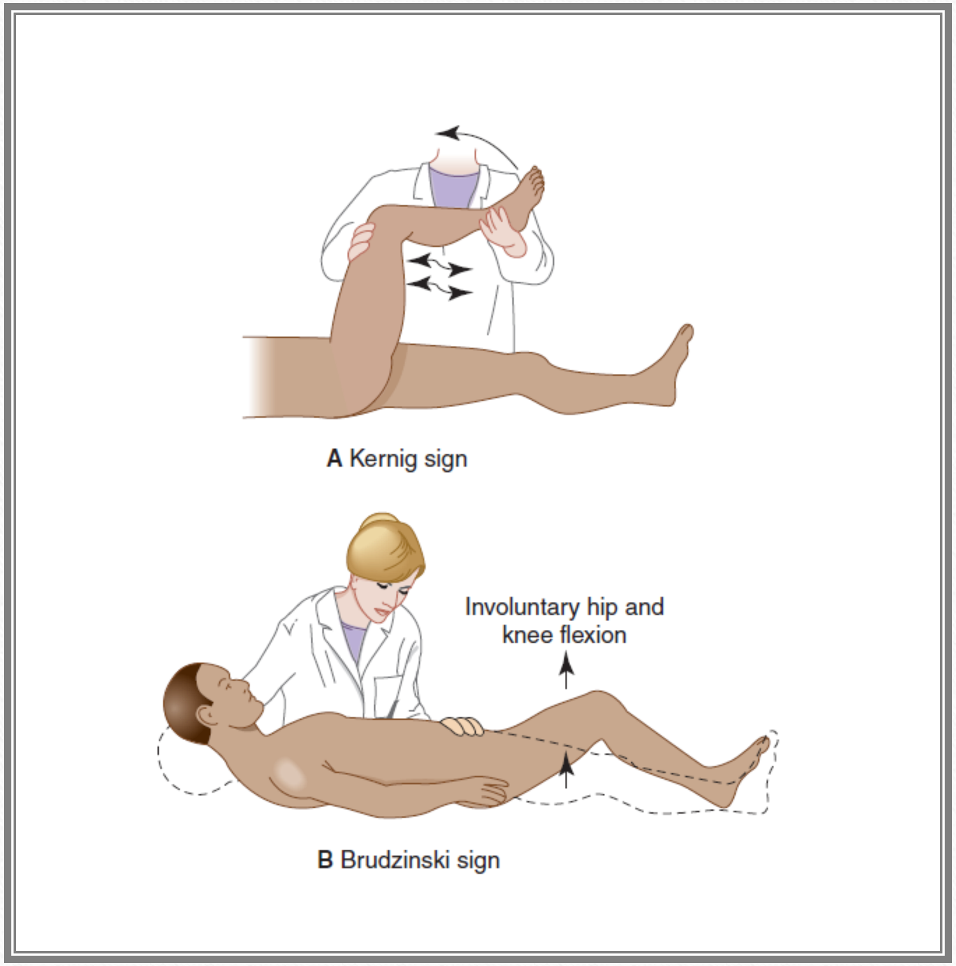
B. Central neurogenic hyperventilation

C. Apneusis

D. Cluster breathing

E. Ataxic breathing





NEUROLOGIC EXAMINATION

- **Signs of Meningeal Irritation**
- Nuchal rigidity, Kernig and brudziniski: They are invaluable in the diagnosis of meningitis or subarachnoid hemorrhage.

- **Pupils**

- **Normal pupils**—Normal pupils are typically 3 to 4 mm
- in diameter (but larger in children and smaller in the
- elderly) and equal in size bilaterally; they constrict briskly and symmetrically in response to light.
- Normally reactive pupils in a comatose patient are characteristic of a metabolic cause
- **Thalamic pupils**—Slightly smaller (~2 mm) **reactive** pupils are present in the early stages of thalamic lesion.

- **Fixed, dilated pupils**—Pupils greater than 7 mm in diameter and **fixed** (unreactive to light) usually result from compression of the oculomotor (III) cranial nerve. <u>**If unilateral think of transtentorial herniation of the medial temporal lobe from a supratentorial mass**</u>. If bilateral think of anticholinergic or sympathomimetic drug intoxication(Not related to oculomotor nerve).
- **Fixed, midsized pupils**—Pupils fixed at approximately 5 mm in diameter are the result of brainstem damage at the midbrain level.

- **Pinpoint pupils** (1-1.5 mm in diameter): Think of opioid overdose, organophosphorus poisoning and pontine lesion (infarction or hemorrhage) .
- Pinpoint pupils may appear unreactive to light except when viewed with a magnifying glass.
- These causes can be distinguished by the administration of naloxone or by the associated defects in horizontal eye movements that usually accompany pontine lesions.
- Other causes: miotic eye drops, or neurosyphilis (Argyll Robertson pupils).

- **Asymmetric pupils**—Asymmetry of pupillary size (anisocoria) with a difference of 1 mm or less in diameter is a normal finding that occurs in 20% of the population.
- If more than 1 mm always think of a structural cause of coma.

- **Fundoscopy**

- Examination of the optic fundi may reveal papilledema or retinal hemorrhages compatible with chronic or acute hypertension, or an elevation in intracranial pressure.
- Subhyaloid (superficial retinal) hemorrhages in an adult strongly suggest subarachnoid hemorrhage.

Papilledema

Disc edema and a large peripapillary subhyaloid hemorrhage

- **Eye Movements**

- **Methods of testing**—In the comatose patient, eye movements are tested by

a. Inspection for spontaneous eye movement.

b. And by stimulating the vestibular system (semicircular canals of the middle ear) by passive head rotation (the **oculocephalic reflex**, or **doll’s-head maneuver**) or using the stronger stimulus of cold water irrigation against the tympanic membrane (**oculovestibular reflex**, or **cold-water caloric** testing)

- **Normal movements**—A comatose patient with intact brainstem function has full conjugate horizontal eye movements, which occur either spontaneously (as "roving eye movements") or during the doll's-head maneuver, as well as tonic conjugate deviation of both eyes toward the side of the ice-water irrigation during cold-water caloric testing.

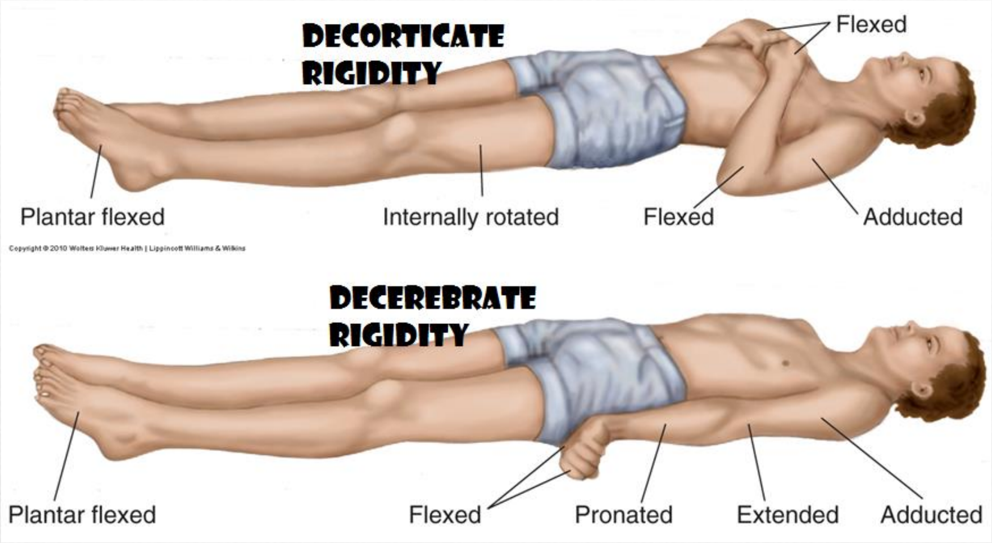
- Full horizontal eye movements in a comatose patient exclude a structural lesion in the brainstem as the cause of coma and suggest a non-structural (eg, metabolic) cause.

- **Motor Response to Pain**

- The motor response to pain is assessed by applying strong pressure on the supraorbital ridge, sternum, or nail beds.
- The response to such stimuli can indicate whether the condition causing coma affects the brain symmetrically (as is typical of metabolic and other diffuse disorders) or asymmetrically (as in unilateral structural lesions).
- The motor response to pain may also help to localize the anatomic level of cerebral dysfunction or provide a guide to the depth of coma.

- With cerebral dysfunction of only moderate severity, patients may localize an offending stimulus by reaching toward the site of stimulation.
- Although such “semipurposeful” localizing responses can be difficult to distinguish from the reflex responses, movements that involve limb abduction almost never represent reflex movements.
- A **decorticate** response to pain (flexion of the arm at the elbow, adduction at the shoulder, and extension and internal rotation of the leg and ankle) is classically associated with lesions that involve the thalamus directly or large hemispheric masses that compress the thalamus from above.

- A **decerebrat**e response (extension at the elbow, internal rotation at the shoulder and forearm, and leg extension) tends to occur when brain dysfunction has descended to the level of the midbrain. Thus decerebrate posturing generally implies more severe brain dysfunction than decorticate posturing
- **Bilateral symmetric posturing** may be seen in both structural and metabolic disorders.
- **Unilateral or asymmetric posturing** suggests structural disease in the contralateral cerebral hemisphere or brainstem.


DECORTICATE
RIGIDITY
Flexed
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Internally rotated
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DECEREBRATE
RIGIDITY
Plantar flexed
Flexed
Pronated
Extended
Adducted

Glasgow Coma Scale

Score	Eye Opening	Verbal Response	Motor Response
1	None	None	None
2	To pain	Vocal but not verbal	Extension
3	To voice	Verbal but not conversational	Flexion
4	Spontaneous	Conversational but disoriented	Withdraws from pain
5	—	Oriented	Localizes pain
6	—	—	Obeys commands

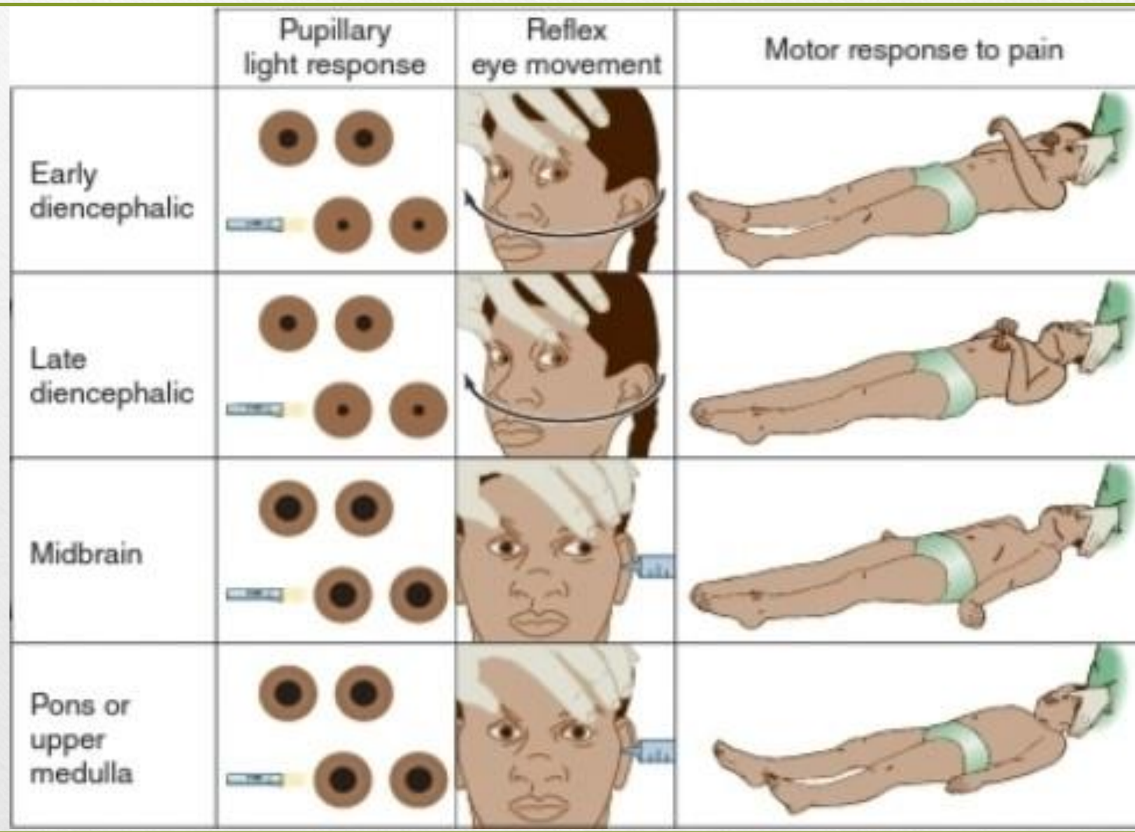
Supratentorial Structural Lesions

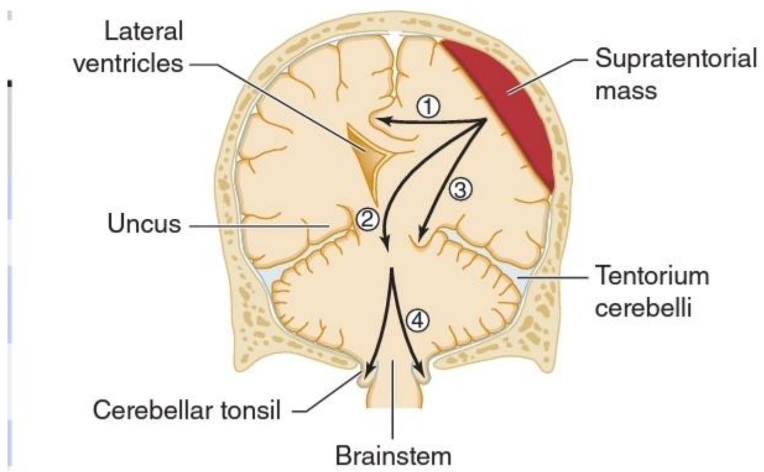
- Symptoms and signs include contralateral hemiparesis, contralateral hemisensory loss, aphasia (with dominant, usually left, hemisphere lesions), and agnosia (indifference to or denial of the deficit, with injury to the nondominant hemisphere).
- As the mass expands (commonly from associated edema), the patient becomes increasingly lethargic due to compression of the contralateral hemisphere or thalamus.
- Stupor progresses to coma, but findings on examination often remain asymmetric.

- With rostral–caudal (downward) progression of brain injury, the thalamus, midbrain, pons, and medulla become sequentially involved, and the neurologic examination reveals dysfunction at successively lower anatomic levels.
- This segmental pattern of rostral-caudal involvement strongly supports the diagnosis of a supratentorial mass with downward transtentorial herniation and dictates the need for neurosurgical intervention.

- At the fully developed midbrain level (midsized, unreactive pupils), chances of survival without severe neurologic impairment decrease rapidly, especially in adults.
- Once the pontine level of dysfunction is reached (unreactive pupils, and absent horizontal eye movements), a fatal outcome is inevitable.
- Supratentorial mass lesions may cause herniation of the medial portion of the temporal lobe (the uncus) over the edge of the cerebellar tentorium.
- This exerts direct pressure on the upper brainstem and produces signs of oculomotor (III) nerve and midbrain compression, such as ipsilateral pupillary dilatation and impaired adduction of the eye **(uncal syndrome)**, which may precede loss of consciousness.

Neurologic signs in coma with downward transtentorial herniation





Anatomic basis of herniation syndromes

- (1) cingulate herniation under the falx,
- 2) downward transtentorial (central) herniation,
- (3) uncal herniation over the edge of the tentorium, or
- (4) cerebellar tonsillar herniation into the foramen magnum.

Subtentorial Structural Lesions

- Coma of sudden onset with focal signs of brainstem dysfunction suggests a subtentorial structural lesion.
- Abnormal pupillary function and impaired eye movement are the findings most suggestive of a subtentorial structural lesion, especially if these abnormalities are **asymmetric**.
- Midbrain lesions cause loss of pupillary function: the pupils are midsized (approximately 5 mm in diameter) and unreactive to light.
- Pontine hemorrhage, pontine infarction, or compression of the pons by adjacent cerebellar hemorrhage or infarction produces pinpoint pupils.

- Brainstem lesions(infratentorial) cause conjugate gaze deviation away from the side of the lesion (and toward a hemiparesis), or disconjugate eye movements such as internuclear ophthalmoplegia.
- Supratentorial lesions cause conjugate gaze deviation toward the side of the lesion and away from the hemiparesis.

Diffuse Encephalopathies

- There are usually no focal signs, such as hemiparesis, hemisensory loss, or aphasia and consciousness is lost after a period of **progressive somnolence or agitated delirium.**
- Usually symmetric neurologic examination is the rule.
- Asterixis, myoclonus, and tremor preceding coma are important clues that suggest metabolic disease.
- Symmetric decorticate or decerebrate posturing can be seen with hepatic, uremic, anoxic, hypoglycemic, or sedative drug–induced coma.

- Reactive pupils in the presence of otherwise impaired brainstem function is the hallmark of metabolic encephalopathy.
- A few metabolic causes of coma can also impair pupillary light reflexes, including massive barbiturate overdose with apnea and hypotension, acute anoxia, marked hypothermia, anticholinergic poisoning (large pupils), and opioid overdose (pinpoint pupils)

Pathophysiologic Assessment of the Comatose Patient

	Supratentorial Structural Lesion	Subtentorial Structural Lesion	Diffuse Encephalopathy/Meningitis
Pupil size and light reaction	Usually normal size (3-4 mm) and reactive; large (>7 mm) and unreactive with transtentorial herniation	Midsized (about 5 mm) and unreactive with midbrain lesion; pinpoint (1-1.5 mm) and unreactive with pontine lesion	Usually normal size (3-4 mm) and reactive; pinpoint (1-1.5 mm) and sometimes unreactive with opiates; large (>7 mm) and unreactive with anticholinergics
Reflex eye movements	Normal (gaze preference toward side of lesion may occur)	Impaired adduction with midbrain lesion; impaired adduction and abduction with pontine lesion	Usually normal; impaired by sedative drugs or Wernicke encephalopathy
Motor responses	Usually asymmetric; may be symmetric after transtentorial herniation	Asymmetric (unilateral lesion) or symmetric (bilateral lesion)	Usually symmetric; may rarely be asymmetric with hypoglycemia, hyperosmolar nonketotic hyperglycemia, or hepatic encephalopathy

PERSISTENT VEGETATIVE STATE

- Some patients who are comatose because of cerebral hypoxia, global cerebral ischemia, head trauma, or bilateral hemispheric strokes regain wakefulness but not awareness.
- If this persists for at least 1 month, it is termed **persistent vegetative state**.
- Such patients exhibit spontaneous eye opening and sleep–wake cycles, which distinguish them from patients in coma, and have intact brainstem and autonomic function.
- However, they neither comprehend nor produce language, and they make no purposeful motor responses and may persist for years.
- Recovery of consciousness from nontraumatic causes is rare after 3 months, and recovery from traumatic causes is rare after 12 months.

LOCKED-IN SYNDROME

- It is caused by a lesion below midpons.
- Such patients appear comatose but are awake and alert although mute and quadriplegic.
- The diagnosis is made by noting that voluntary eye opening, vertical eye movements and ocular convergence.
- During the examination of any apparently comatose patient, the patient should be told to “open your eyes,” “look up,” “look down,” and “look at the tip of your nose” to elicit such movements.
- Sleep wake cycle and EEG are normal and outcome is variable and related to the underlying cause.

End